Machine Learning Ensemble Model for Stock Trading Signal Generation

C964 - Computer Science Capstone

Western Governors University

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Task 2 parts A, B, C and D

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LETTER OF TRANSMITTAL

February 10, 2025

Alex Carter, CTO
Global Investment Solutions
123 Finance Avenue
New York, NY

Subject: Request for Approval to Develop the Ensemble Model Investment Platform

Dear Mr. Carter,

I am writing to request your approval for an innovative new revenue source that addresses the financial challenges faced by Americans living paycheck-to-paycheck. As highlighted in the project proposal, factors that limit the ability of most Americans to grow their wealth include escalating living costs, time constraints, and limited exposure to investing.

Summary of the Problem
A massive segment of potential investors lacks both the resources and time to research viable opportunities. Even with tools such as Robinhood and Acorns, the barriers to understanding and capitalizing on market trends remain high.

Proposed Solution
The Ensemble Model (EM), powered by Extreme Gradient Boosting (XGBoost), is designed to generate daily buy/sell signals based on stock price movements. By automatizing data analysis, it simplifies investment decisions and lowers entry hurdles for individuals who lack time or expertise.

Background
My background includes a Computer Science degree and seven years of analytical project management experience, where I have managed projects valued in the hundreds of thousands and overseen aggregate budgets in the millions annually.

Benefits to Global Investment Solutions

1. **Expand Market Access:** Attract a new demographic of customers who wish to invest but lack the familiarity.
2. **Strengthen Brand Image:** Position the company as a champion of financial inclusion and empowerment.
3. **Foster Sustainable Growth:** Promote upselling opportunities as users gain confidence and explore broader offerings.

Costs, Timeline, and Data
• Total development costs for the proof-of-concept range from $1,625 to $2,600 due to a lean open-source stack.
• The project will follow an Agile methodology for rapid iterations, offering feedback and improvements

throughout short, time-boxed sprints.
• The data is sourced publicly from Yahoo Finance, ensuring no privacy issues. Model security measures will safeguard our intellectual property.

Kindly review this request and consider authorizing our team to proceed with developing the Ensemble Model. We anticipate that this lightweight pilot project will set the stage for a revolutionary, inclusive investment platform.

Thank you for your time and support. I am available to discuss any details or answer any questions you may have.

Sincerely,

Timothy Javins
Project Manager
Global Investment Solutions
timothy.javins@gis.ai

Project Proposal

Problem

With the rising costs of living, unattainable single-family house prices, the increasing wealth chasm between rich and poor, and the ever-busier American lifestyle, Americans need a new entry point to wealth development. Though innovative companies, such as Robinhood and Acorns, have lowered the difficulty for Americans to invest, people still lack the time to investigate opportunities and actively participate during the work week. Americans need to make data-driven investment decisions but lack the ability to collect the data, much less analyze it.

LendingClub and PYMNTS.com published *New Reality Check: The Paycheck-To-Paycheck Report* in March 2024, which highlighted how rising costs and economic shifts are impacting people's ability to save and invest. Most importantly, the report revealed that **62% of Americans live paycheck-to-paycheck**. Fusing this data with population data from the US Census Bureau, there are over 211 million people who would benefit from a solution to this problem, including over 45 million children. This means there are more than 166 million adults who may resonate with small-scale investment opportunities. (U.S. Census Bureau, 2024)

The US Census Bureau lists about 78 million households (59%) earning less than $100,000 per year. In estimating the **annual buying power** of the paycheck-to-paycheck group, it would be simple to assume the lowest income bracket and derive from that demographic. However, an estimate of **$9.233 *trillion***--based on 80,642,000 households and the average income of $114,500 (U.S. Census Bureau, 2024)--is supported by the fact that the *New Reality Check* included 48% of *individuals* earning *over $100,000* annually who are living paycheck to paycheck. Notably, there was also a significant number who expressed interest in investing but felt barred by high entry points or lack of knowledge. This emphasizes the importance of creating accessible investment opportunities that cater to those starting with limited resources.

Younger generations, especially Millennials and Gen Z, are particularly intriguing here. They are tech-savvy, comfortable with apps, and open to non-traditional financial avenues. Even with limited funds, they show a keen interest in micro-investing platforms that allow them to start with as little as a dollar. This taps into a desire many have to take control of their financial destinies, even when starting small.

Still, many people hesitate to invest because it feels intimidating, complicated, or unfamiliar. **Global Investment Solutions does not currently present a low-cost offering for the cash-strapped, timebound, unsophisticated majority of American would-be investors.** While not investment savvy, these managers of trillions of dollars are often tech-savvy and find advanced analytics enticing. Providing a head start and simplifying investment processes could empower them to get started.

Solution

By zeroing in on this demographic, this proposed project is tapping into a movement of people ready to grow, learn, and invest in their futures—even if only starting with a few dollars. GIS can enable investment for Americans who are juggling tight budgets but are eager to dip their toes into investing, even if starting with spare change. Some of these individuals earn between $20,000 and $50,000 annually. Though not swimming

in disposable income, they are collectively controlling trillions of dollars and the spark is there to grow their financial future, bit by bit. This is an entirely new market for GIS to enter.

To distribute financial intelligence to the masses, the solution needs to be copyable. To be copyable, it needs to be artificial. By training a machine learning algorithm on a given stock's performance over time, a machine can predict whether a stock price will rise or fall. This is the bare minimum of knowledge that an investor needs. By providing quality signals like this daily, GIS stands to not only open up a vast market but also contribute to greater financial inclusion and empowerment.

The Ensemble Model (EM), proposed now, will be an early-stage example of the power of mathematically modeling stock performance. This model will provide buy/sell signals that will be understood without a background in finance. Anyone subscribed to the product will be able to begin investing with a jumpstart, making trades with an easy buy/sell signal. These new GIS customers will be able to generalize to other GIS products and investment vehicles over time, as their wealth grows and their intuition develops.

Deliverables

The Ensemble Model will be written in the open-source programming language Python, utilizing the Extreme Gradient Boosting (XGBoost) library for creating and training an ensemble of decision tree-based models. It will generate daily trade signals, which can be integrated into the final User Experience-optimized product.

A line graph will chart the performance of a trading strategy following the model's signals over time. For comparison, the same line graph will also chart a simple buy and hold strategy applied to a Standard & Poor's 500 Index (S&P 500) exchange-traded fund (ETF). This will answer the question, "Can this strategy outperform the S&P 500?"

A bar graph will display various performance metrics comparing the model's strategy to the S&P 500. These metrics will be well-known indicators, including Calmar, Sharpe, and Sortino ratios, annualized returns, and maximum drawdown.

For context, candlestick charts will show daily summary price data, including open, close, high, low, range, and direction.

Data Summary

The data used to train the models will be sourced from Yahoo Finance, including daily price data such as:

- High (floating point decimal)
- Low (floating point decimal)
- Open (floating point decimal)
- Close (floating point decimal)
- Volume (floating point decimal)
- Date (text)

Because the project will rely exclusively on publicly available data sourced from Yahoo Finance, which contains no personally identifiable information (PII), there are no immediate concerns regarding PII or customer data

privacy. All intellectual property developed in the project will be safeguarded in accordance with existing policies and best practices.

The raw stock data will be enriched through a feature engineering process that extracts key predictors for the ensemble model. The preprocessed data will be transformed by computing the daily return as the percentage change of the closing price, the volatility as the 21-day rolling standard deviation of these returns, and the momentum as the 21-day rolling average of the closing price. These engineered features will capture essential trends and fluctuations, and after dropping any missing values, they will form a clean dataset that serves as the foundation for training the Ensemble Model.

Once fully developed, the product will pull fresh data after the closing bell each day, saving the data in a secure location local to the Ensemble Model. However, the initial evaluation model will use static timeframes.

The data will be time-series data, limited to summary metrics for each day. This simple daily time interval will make for an easily understood entry point for new investors, but will have other benefits as well. A finer granularity, such as price changes by the minute or by the second, would present higher costs and information overload for the user. To avoid these complications and others resulting from potential pattern day trading (PDT), more granular data and higher frequency signals are out of scope for this project.

Objective and Hypothesis

The primary objective of this project is to establish a baseline proof-of-concept in support of developing a low-cost, accessible investment platform that empowers Americans, especially those living paycheck-to-paycheck, to make data-driven investment decisions. By utilizing machine learning algorithms, the platform will aim to simplify the investment process and provide users with actionable buy/sell signals. This will help users, particularly those with limited financial knowledge and resources, to start investing with confidence and gradually build their wealth.

The hypothesis is that the Ensemble Model can generate reliable buy/sell signals that outperform traditional investment strategies, such as a simple buy and hold approach, by utilizing an XGBoost machine learning model to predict stock price movements. By providing users with accurate and timely investment recommendations, the platform will not only improve their financial outcomes but also increase their confidence and participation in the stock market. This, in turn, will contribute to greater financial inclusion and empowerment for a broader demographic, ultimately leading to higher customer satisfaction and increased revenue for Global Investment Solutions.

Project Methodology

Instead of following a strict, sequential process, the project will benefit from an Agile approach, which embraces an iterative and incremental cycle. This will allow for building, testing, and refining the investment platform continuously in close collaboration with stakeholders. Below is an outline of the Agile process for the project:

1. **Backlog Creation & Prioritization**

    - Collaborate with key stakeholders (target users, product managers, and technical leads) to gather user stories and define acceptance criteria.

- Create and maintain a prioritized product backlog that captures functional requirements, enhancements, and potential technical improvements.
- Continuously update and refine the backlog based on new insights, user feedback, and market changes.

2. **Sprint Planning**

    - Organize the work into short, time-boxed iterations (sprints), typically lasting two to four weeks.
    - Select a set of user stories from the backlog that can be feasibly developed, tested, and delivered within the sprint.
    - Define clear sprint goals and outline tasks, ensuring that each chosen story has well-defined criteria for completion.

3. **Iterative Development & Testing**

    - Develop features in small increments through cross-functional teams that collaborate on design, coding, and immediate testing.
    - Integrate unit tests, functional tests, and user acceptance tests into the development process.
    - Leverage continuous integration to ensure that new code is consistently merged, built, and tested, reducing integration issues and enabling early detection of defects.

4. **Sprint Review & Demo**

    - At the end of each sprint, demonstrate the working product increment to stakeholders.
    - Gather feedback on new features, usability, and performance against the defined acceptance criteria.
    - Discuss any required adjustments and incorporate this feedback into the backlog for upcoming sprints.

5. **Sprint Retrospective & Continuous Improvement**

    - Conduct team retrospectives to analyze the sprint process, identify successes and challenges, and propose actionable improvements.
    - Adjust processes and practices to enhance productivity, collaboration, and quality in future sprints.
    - Use the lessons learned from each retrospective to streamline workflow and better align the product with user needs.

6. **Ongoing Release & Maintenance**

    - As product increments reach a sufficiently refined state, prepare for regular releases that provide value to end users.
    - Monitor performance, gather continuous user feedback, and address bugs or enhancement opportunities through additional sprints.
    - Ensure the platform evolves with market demands and user expectations, sustaining an agile response to change over time.

Timeline

Milestone	Duration	Projected Start Date	Anticipated End Date
Backlog Creation & Prioritization	2 hours	Feb. 11, 2025	Feb. 11, 2025
Sprint Planning	2 hours	Feb. 11, 2025	Feb. 11, 2025
Iterative Development & Testing (Sprint 1)	4 hours	Feb. 12, 2025	Feb. 12, 2025
Sprint Review & Demo (Sprint 1)	1 hours	Feb. 13, 2025	Feb. 13, 2025
Sprint Retrospective & Continuous Improvement (Sprint 1)	2 hours	Feb. 13, 2025	Feb. 13, 2025
Iterative Development & Testing (Sprint 2)	4 hours	Feb. 14, 2025	Feb. 14, 2025
Sprint Review & Demo (Sprint 2)	1 hours	Feb. 17, 2025	Feb. 17, 2025
Sprint Retrospective & Continuous Improvement (Sprint 2)	2 hours	Feb. 17, 2025	Feb. 17, 2025
Ongoing Release & Maintenance	Ongoing	Feb. 18, 2025	Ongoing

Evaluation Plan

During the initial phase, each module—whether it be data collection, preprocessing, or feature engineering—will be subjected to unit tests to ensure that individual functions are performing as expected. As new features are created (for instance, those generated by the feature engineering process), unit testing will be employed to verify that the transformations correctly handle missing values and produce a clean dataset.

Once these modules are integrated, continuous integration will be used to merge, build, and test the complete codebase, reducing integration issues.

Subsequent integration testing will verify whether the interaction between modules (such as between data preprocessing, model training, and strategy implementation) is seamless.

At the end of each sprint, system testing will be carried out by simulating real-use cases. This will include backtesting the trading strategy by using historical data. The team will also perform user acceptance testing with stakeholders to ensure the product meets the defined requirements and usability expectations.

Finally, upon project completion, the overall validation will be conducted by comparing key performance metrics—including Mean Squared Error, Annualized Return, and Sharpe, Sortino, and Calmar ratios—against established benchmarks. These metrics will serve as objective measures to confirm that the Ensemble Model reliably produces buy/sell signals and meets the hypothesis that it can outperform certain traditional investment strategies.

This comprehensive evaluation plan ensures all aspects of the product will be robustly tested and validated before deployment.

Funding Requirements

For establishing the Ensemble Model proof-of-concept, the funding requirements are kept to a minimum by leveraging open-source tools and a lean development process. Key funding elements include:

- **Development Resources:**
  A significant portion will cover the developer's time. For developing the EM, an estimated 25 to 40 labor hours at an estimated rate of $65 per hour results in a budget range of $1,625 to $2,600.

- **Infrastructure & Tools:**
  Since the project utilizes open-source libraries (e.g., Python, XGBoost) and local-to-the-model data storage during development, infrastructure costs are minimal. The project will piggyback on mainline business infrastructure, including for version control and continuous integration pipelines. However, there is a potential cost in cloud hosting if needed for scalability tests.

- **Contingency & Miscellaneous Costs:**
  A small contingency reserve (around 10-15% of the total estimated costs) will be set aside to address unforeseen expenses such as additional testing tools or minor cloud computing costs.

This modest budget supports an agile, iterative approach, allowing us to rapidly develop and refine the proof-of-concept before considering investment in a full-scale, production-ready platform.

The Application

Please view the Ensemble Model notebook to view the application in detail.

https://colab.research.google.com/drive/1TGX4S4OOT64sEsBF9D-wbFg0JTZfXgUw?usp=sharing

Post-Implementation Report

Stakeholder Impact

This project created value for all stakeholders. End users—particularly those living paycheck-to-paycheck—stand to benefit from an accessible, low-cost investment platform that simplifies data-driven decision-making and lowers the entry barrier to investing. By delivering actionable buy/sell signals, the model proved the potential to empower users to make investment choices, ultimately improving financial outcomes.

For internal stakeholders, such as executives, product managers, and technical leads, the iterative Agile process ensured continuous improvement and swift adaptation to feedback, fostering innovation and operational efficiency. Overall, the project proved not only the potential to enhance customer satisfaction and financial inclusion but also to position Global Investment Solutions to capture new market opportunities and drive sustainable growth.

Solution Summary

The Ensemble Model showed that it is possible to bridge the gap between raw financial data and investor insights by orchestrating end-to-end data management—from data pulling and preprocessing to model training and prediction. This approach ensured technical feasibility while addressing the business need for a simple yet effective investment decision tool.

The Ensemble Model implemented an end-to-end pipeline:

- It downloaded raw stock and S&P 500 data from Yahoo Finance.
- It performed data preprocessing (dropping NaNs, converting types, engineering features like returns, volatility, and momentum).
- It trained an XGBoost regression model to predict the following day's return.
- It backtested the trading strategy and computed key performance metrics, such as Mean Squared Error, Sharpe, Sortino, and Calmar ratios.

This full-cycle approach provided an accessible investment signal generation and validation system, addressing both the problem of information overload and the need for actionable insights.

To use test the Ensemble Model, open the notebook and run all the cells. To run the entire notebook at once, choose "Runtime" from the menu and then "Run all" or press Ctrl + F9.

To run cells individually, press the play button in the top left corner of the cell or press Ctrl + Enter.

To input your own date range for testing the model, change the values of the `start_date` and `end_date` variables in the **Test the Model** section, matching the "YYYY-MM-DD" date format. After inputting the dates, run the cell and all subsequent cells.

Data Summary

Data Governance & Privacy Considerations

The application pulled its data from API calls and stored it local to the model for further processing. The project relied exclusively on publicly available data sourced from Yahoo Finance API calls, which contained no personally identifiable information, so there were no concerns regarding PII or customer data privacy.

However, the project team considered future development advances and prepared the deliverables to be safeguarded in the proprietary aspects of the model, including its configurations, training data, and performance metrics. The team prepared measures such as secure storage, regular access audits, and proper version control to be implemented, ensuring that intellectual property and model integrity remain protected against unauthorized access or misuse. These precautions helped maintain data governance standards and secure the future transition from a proof-of-concept to a production-ready platform.

Collection: The application downloaded stock data for tickers such as AAPL (Apple) and ^GSPC (S&P 500) within defined date ranges.
Preprocessing: The application read and cleaned the data, dropping missing values and specific rows. Some columns were renamed/retyped.
Feature Engineering: The application created new columns, such as Return, Volatility, and Momentum, and computed new values for these columns in order to capture the key predictors.
Management: The application saved processed data at multiple stages as CSV files, providing a layer of versioning through the design, development, and testing phases.

Machine Learning

Method Employed:
XGBoost Regression Model

What:
The Extreme Gradient Boosting (XGBoost) model was used to predict the next day's stock return based on features such as Volatility and Momentum.

How:
The model was trained via the following process.

The engineered features were split into training and test sets to evaluate the model's performance effectively. This process involved dividing the dataset into two distinct subsets:

> **Training Set:** This subset was used to train the XGBoost model. It included a portion of the data with known outcomes, allowing the model to learn the relationships between the features (such as Volatility and Momentum) and the target variable (next day's stock return). By exposing the model to a variety of data points, it could identify patterns and make accurate predictions.
>
> **Test Set:** This subset was used to evaluate the model's performance after training. It consisted of data that the model had not seen during the training phase. By assessing the model's predictions on this

unseen data, we could gauge its ability to generalize and perform well on new, real-world data. The test set provided an unbiased evaluation of the model's accuracy and helped identify any overfitting issues.

Splitting the data into training and test sets ensured that the model's performance metrics, such as Mean Squared Error, reflected its true predictive capabilities and not just its ability to memorize the training data.

The model was constructed with an objective of `reg:squarederror`, which means it aimed to minimize the squared differences between the predicted and actual values, a common objective for regression tasks. This objective function helps the model focus on reducing large errors, leading to more accurate predictions.

Additionally, the model was trained with a high number of estimators, meaning it used a large number of boosting rounds to iteratively improve the model's performance. Each estimator in the boosting process corrected the errors of the previous ones, allowing the model to capture complex patterns and relationships in the data. This approach enhanced the model's ability to generalize and make accurate predictions on unseen data.

After the XGBoost model was trained, it was used to make predictions on the test set. These predictions represented the model's forecast of the next day's stock returns based on the features it had learned during training. The accuracy and effectiveness of these predictions were then evaluated using performance metrics, with Mean Squared Error (MSE) being a primary metric.

Predictions: The model generated predicted values for the target variable (next day's stock return) for each data point in the test set. These predictions were compared to the actual observed values to assess the model's accuracy.

Evaluation: The performance of the model was quantified using Mean Squared Error (MSE). MSE measures the average squared difference between the predicted values and the actual values. It is calculated as follows:

```
from sklearn.metrics import mean_squared_error

mse = mean_squared_error(actual_values, predicted_values)
```

Where `actual_values` are the true stock returns from the test set, and `predicted_values` are the returns predicted by the model.

Mean Squared Error (MSE): MSE is a widely used metric for regression tasks. It penalizes larger errors more than smaller ones, making it sensitive to outliers. A lower MSE indicates that the model's predictions are closer to the actual values, signifying better performance. The formula for MSE is:

$$\text{MSE} = \frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2$$

Where n is the number of data points, y_i are the actual values, and $\hat{y}_i$ are the predicted values.

By computing the MSE, the model's predictive accuracy was quantitatively assessed, providing a clear measure of how well the model performed in predicting stock returns. This evaluation helped identify areas for improvement and guided the determination of whether the model was reliable for making investment decisions.

Why:
XGBoost was chosen because it is a powerful ensemble technique suitable for time series predictions. Its ability to model complex, non-linear relationships in financial data helped generate actionable buy/sell signals, making it an appropriate choice for the project's goal of outperforming traditional strategies.

After the proof-of-concept model is validated, development teams could also explore community-driven investment models or social features that would amplify engagement. People often feel more confident and motivated when they can share experiences and progress with peers. Gamification elements might also make investing feel more accessible and enjoyable.

The model will need to update its training frequently, perhaps monthly or weekly. To demonstrate why, two versions of the model are provided with different performance time horizons.

Validation

The model's performance was validated using MSE and established investment performance metrics.

Backtesting Performance Metrics

Backtesting Performance Metrics:
The model's performance was further validated using several key financial metrics during the backtesting phase. These metrics provided a comprehensive evaluation of the trading strategy's effectiveness compared to the S&P 500 benchmark:

Sharpe Ratio:
The Sharpe Ratio measures the risk-adjusted return of the trading strategy. It is calculated by dividing the excess return (return above the risk-free rate) by the standard deviation of the returns. A higher Sharpe Ratio indicates better risk-adjusted performance. The formula is:

$$\text{Sharpe Ratio} = \frac{R_p - R_f}{\sigma_p}$$

Where R_p is the portfolio return, R_f is the risk-free rate, and σ_p is the standard deviation of the portfolio return.

Sortino Ratio:
The Sortino Ratio is similar to the Sharpe Ratio but focuses only on downside risk (negative returns). It is calculated by dividing the excess return by the downside deviation. This metric provides a more accurate measure of risk-adjusted performance for strategies that aim to minimize losses. The formula is:

$$\text{Sortino Ratio} = \frac{R_p - R_f}{\sigma_d}$$

Where σ_d is the downside deviation.

Calmar Ratio:
The Calmar Ratio measures the risk-adjusted return by comparing the annualized return to the maximum drawdown (largest peak-to-trough decline). A higher Calmar Ratio indicates better performance relative to the risk of significant losses. The formula is:

$$\text{Calmar Ratio} = \frac{\text{Annualized Return}}{\text{Maximum Drawdown}}$$

Annualized Return:
The Annualized Return represents the geometric average of the returns over a year, providing a standardized measure of the strategy's performance. It allows for comparison with other investments and benchmarks like the S&P 500. The formula is:

$$\text{Annualized Return} = \left(\prod_{i=1}^{n}(1+R_i)\right)^{\frac{252}{n}} - 1$$

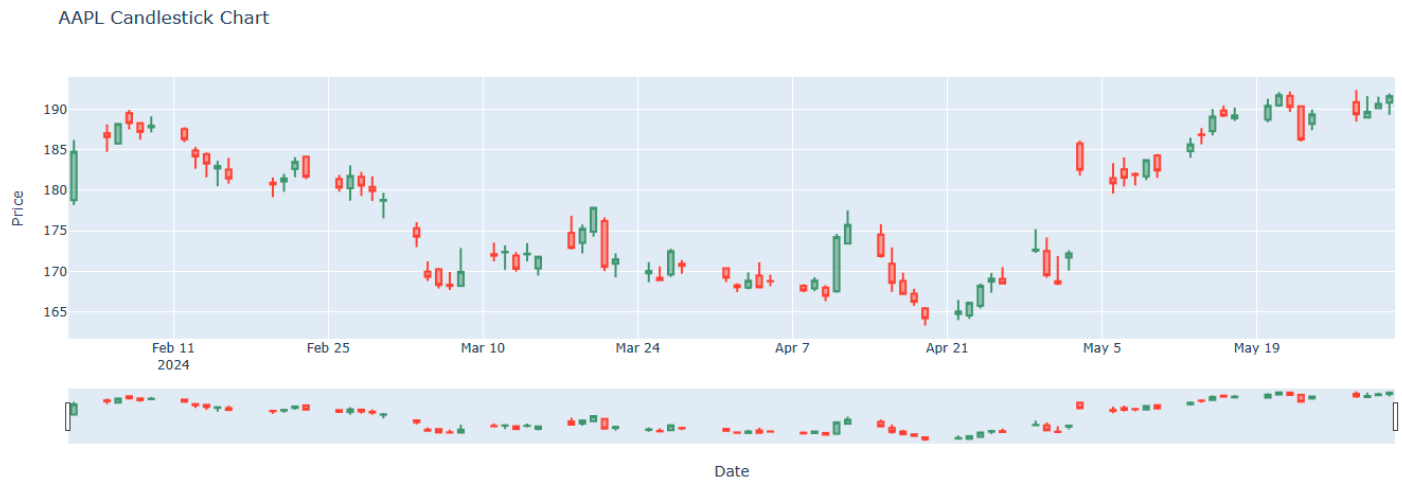
Where R_i are the daily returns, and n is the number of trading days.

These metrics were calculated and compared against the S&P 500 benchmark to evaluate the trading strategy's performance. The model's ability to generate risk-adjusted returns was assessed by using these comprehensive performance metrics. This thorough validation process provided a sober degree of confidence in the model's real-world applicability and robustness.

Visualizations

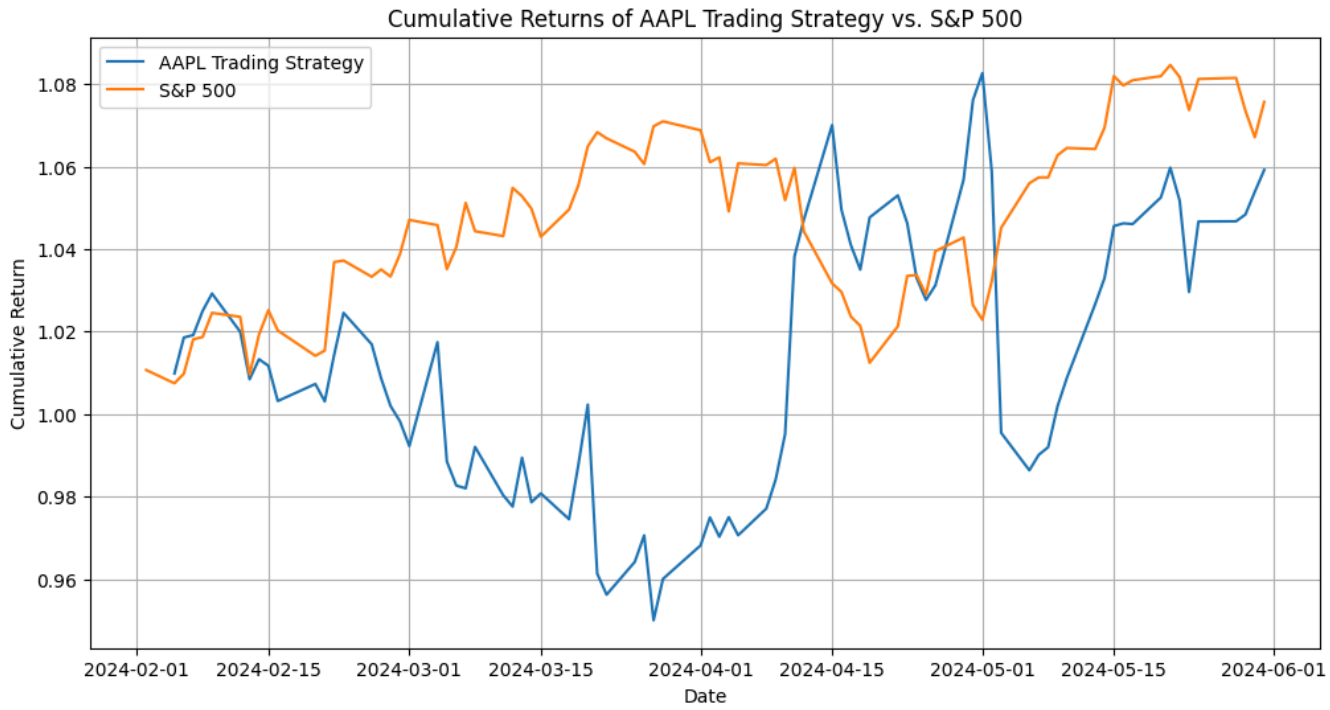
The application generated several visualizations in its final phase.

Candlestick Charts: Using the `plotly` library, one chart was generated for the Apple stock and another for the S&P 500.

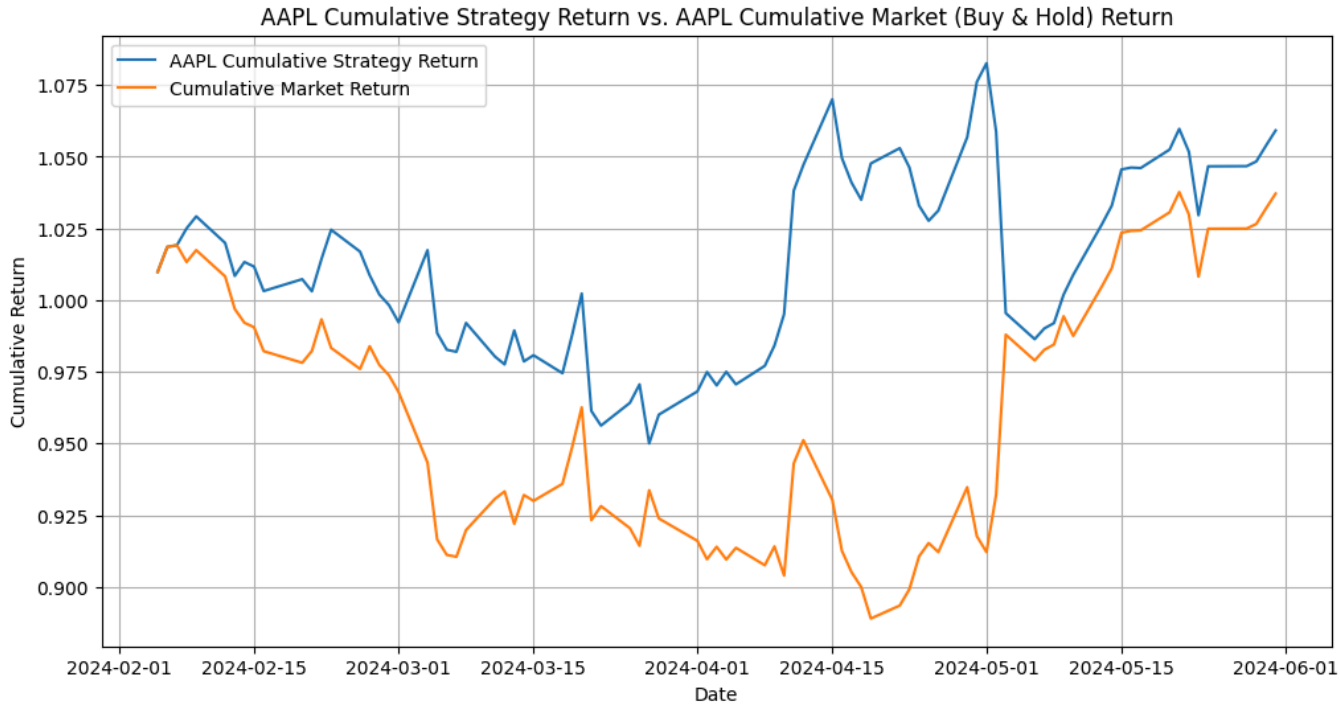


These two interactive charts (pictured above) showed the price movements over the testing period, which provided context for the real-world performance of these two investment vehicles.

Line Charts: Two line charts showed the comparative cumulative returns, generated with the `matplotlib` library.

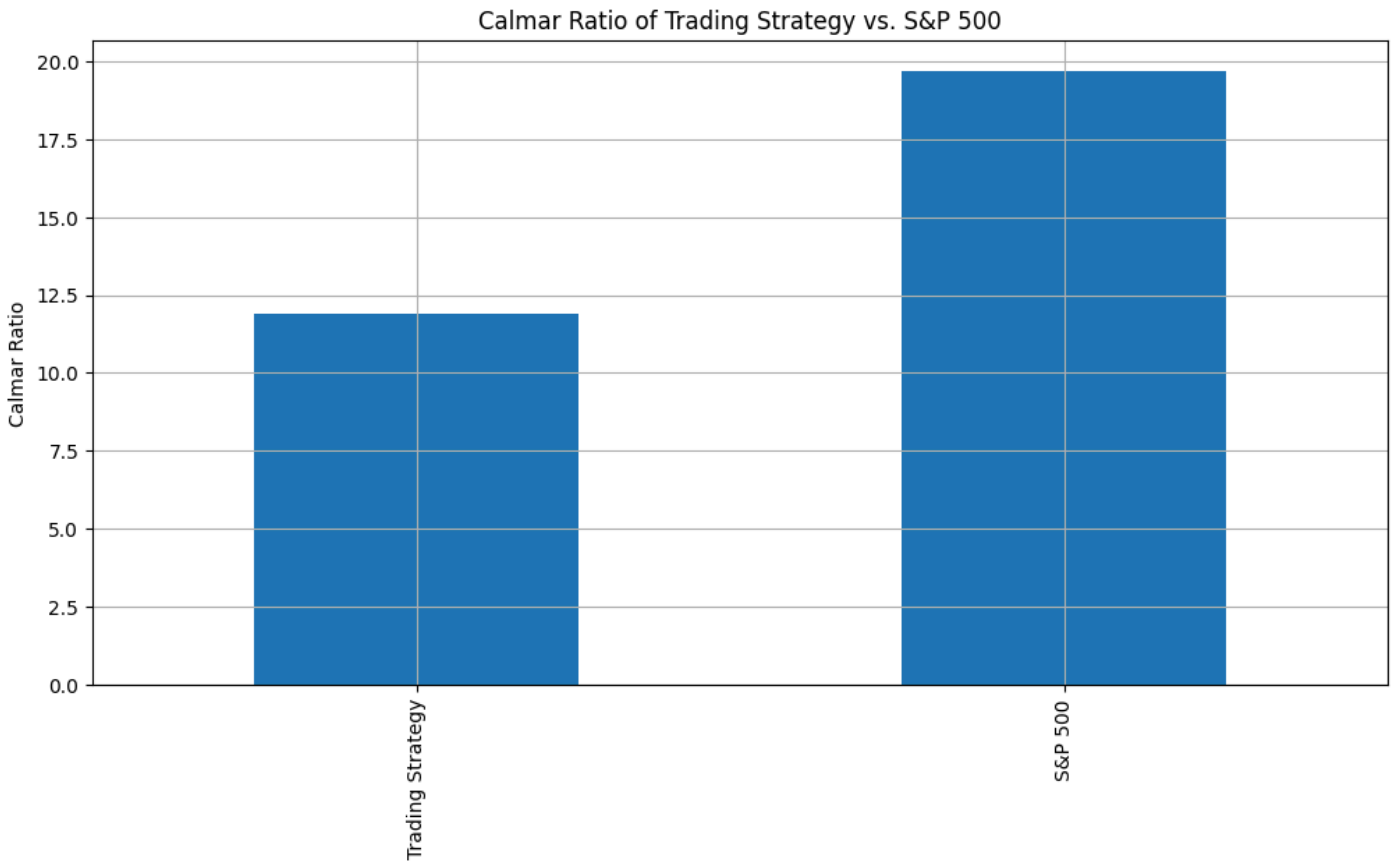


The first chart visualized the trading strategy's performance in comparison to the S&P 500 index. While the S&P 500 outperformed the EM, there were some periods in which the EM outperformed the index.

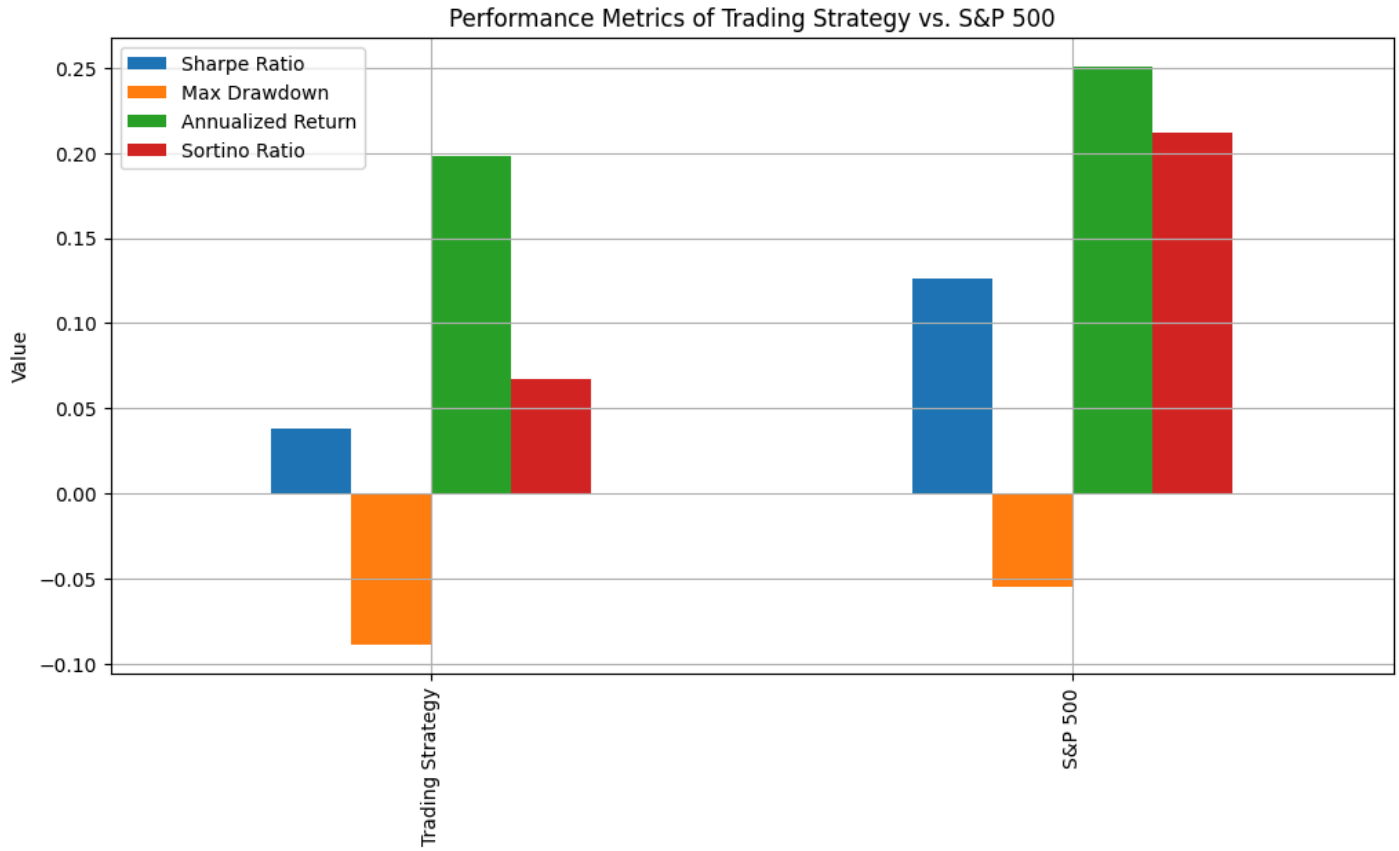


The second chart was similar but compared the cumulative strategy return on Apple stock versus the cumulative market return—reflective of a buy-and-hold strategy—on Apple stock. This chart was meaningful as it proved the model generated performant trading signals and outperformed the Apple stock's market return.

Bar Charts: The application used the `plotly` library to generate two bar charts, each representing the application's generated performance metrics.



The first bar chart showed the Calmar Ratios for the trading strategy and the S&P 500 over the testing period. The higher Calmar ratio of the S&P 500 reflected the less volatile nature of the diversity in the index: the stocks of the top 500 companies. Naturally, the single-stock strategy employed by the model in this proof-of-concept was more volatile, resulting in a lower Calmar ratio.



The second bar chart showed the other performance metrics for each strategy, including Sharpe Ratio, MDD, Annualized Return, and Sortino Ratio. All four metrics showed investing in S&P 500 ETFs was a less volatile, more profitable strategy. For example, the MDD was larger in the Apple strategy and the Annualized Return was smaller. This difference was reflective of the Calmar Ratio and highlighted the advantage of diversification in the S&P 500 Index.

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